Name_____ Date_____ Period_____

LESSON 3-7 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Customary vs. Metric Conversions

Maryland Standard 6.RP.A.3d · Learning target – I can convert a measurement between the customary and metric systems using a given rate.

REMEMBER

- Use the rate you are given, such as 1 in = 2.54 cm, 1 mi ≈ 1.6 km, 1 kg ≈ 2.2 lb.
- Converting to a **smaller** unit: multiply. Converting to a **larger** unit: divide.
- Write **≈** (about) when the rate is rounded.

WORKED EXAMPLE

A hiking trail is 12 kilometers long. About how many miles is that? (1 mi ≈ 1.6 km)

Step 1 A mile is longer than a kilometer, so there are fewer miles. Divide.

Step 2 12 ÷ 1.6 = 7.5

Answer **about 7.5 miles**

WORD BANK

≈ – about equal to

kilogram (kg) – heavier than a pound: 1 kg ≈ 2.2 lb

kilometer (km) – shorter than a mile: 1 mi ≈ 1.6 km

reasonable – the size of the answer makes sense

PRACTICE

show your work · label units

1. Convert. Use the rate given.

a. 6 in (1 in = 2.54 cm) = _____ cm

b. 25 mi (1 mi ≈ 1.6 km) ≈ _____ km

c. 11 lb (1 kg ≈ 2.2 lb) ≈ _____ kg

d. 3 gal (1 gal ≈ 3.8 L) ≈ _____ L

e. 50 kg (1 kg ≈ 2.2 lb) ≈ _____ lb

f. 8 km (1 mi ≈ 1.6 km) ≈ _____ mi

g. 4.4 lb (1 kg ≈ 2.2 lb) ≈ _____ kg

h. 10 mi (1 mi ≈ 1.6 km) ≈ _____ km

2. Complete the ratio table. (1 kg ≈ 2.2 lb)

Step 1 Kilograms to pounds: multiply by 2.2.

Step 2 Pounds to kilograms: divide by 2.2.

Kilograms	1	5		25
Pounds	2.2		22	

3. **Road Sign.** A road sign says the next city is **96 km** away. About how many **miles** is that? (1 mi ≈ 1.6 km)

Step 1 A mile is longer than a kilometer, so the answer is smaller.

Step 2 Divide 96 by 1.6.

Show your work.

Answer_____ miles

4. **Suitcase.** An airline allows a suitcase of up to 23 kg. Ella's suitcase weighs 55 lb. (1 kg ≈ 2.2 lb)

Step 1 Divide 55 by 2.2 to get kilograms.

Step 2 Compare your answer with 23 kg.

a. About how many kilograms does the suitcase weigh? _____ kg

b. Is it under the limit? By how much is it over or under? _____

5. **Compare.** Which race is longer: a 100-meter race or a 100-yard race? (1 m ≈ 1.09 yd) Explain.

Frame *100 m ≈ ___ yd, so the ___ race is longer because ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Find a label that shows both systems (like 1 lb (454 g)). Check that the two amounts make sense.

Name_____ Date_____ Period_____

LESSON 3-7 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Multi-Step System Conversions and Dosage

Maryland Standard 6.RP.A.3d · Learning target – I can solve a multi-step problem that requires converting units before using a rate.

REMEMBER

- **First** convert to the unit the rate uses. **Then** apply the rate.
- Keep track of the units at every step. Label your final answer.

WORKED EXAMPLE

A dog weighs 55 lb. The dose is 6 mg per kg. (1 kg ≈ 2.2 lb) Find the dose.

Step 1 Convert to kilograms: 55 ÷ 2.2 = 25 kg

Step 2 Apply the rate: 25 × 6 = 150

Answer **150 mg**

WORD BANK

dose – the amount of medicine to give

per – for each

multi-step – do one step, then the next

PRACTICE

show your work · label units

1. Complete each step. (1 kg ≈ 2.2 lb; 1 tbsp = 3 tsp; 1 L = 1,000 mL)

a. 77 lb ≈ ____ kg

b. 35 kg × 4 mg per kg = ____ mg

c. 2 tbsp = ____ tsp

d. 1.5 L = ____ mL

2. **Cat Medicine.** A cat weighs 8.8 lb. The dose is 5 mg per kg. (1 kg ≈ 2.2 lb)

Step 1 Divide 8.8 by 2.2 to get kilograms.

Step 2 Multiply the kilograms by 5 mg.

a. Weight in kilograms: _____ kg

b. Dose: _____ mg

3. **Track Practice.** Rosa runs 4 laps of a 400-meter track each day for 5 days. (1 mi ≈ 1.6 km)

Step 1 Multiply 4 × 400 × 5 to get meters.

Step 2 Divide by 1,000 for kilometers.

Step 3 Divide the kilometers by 1.6 for miles.

a. Total meters: _____ m

b. Total kilometers: _____ km

c. About how many miles? _____ miles

4. **Medicine Bottle.** A bottle holds 120 mL of medicine. One dose is 2 teaspoons. (1 tsp ≈ 5 mL)

Step 1 Multiply 2 × 5 mL for one dose.

Step 2 Divide 120 mL by the mL in one dose.

a. About how many milliliters are in one dose? _____ mL

b. About how many doses are in the bottle? _____ doses

5. **Find the Error.** A goat weighs 66 lb. The vitamin dose is 3 mg per kg. Nia multiplied 66 × 3 = 198 mg. Explain her mistake and find the correct dose. (1 kg ≈ 2.2 lb)

Frame *Nia used ___ instead of ___. The correct dose is ___ mg.*

TEST PRACTICE

choose the best answer

6. A dog weighs 33 lb. The dose is 2 mg per kg. (1 kg ≈ 2.2 lb) What is the dose?

A 30 mg **B** 66 mg **C** 145.2 mg **D** 15 mg

7. A soup recipe needs 3 quarts of broth. About how many liters is that? (1 qt ≈ 0.95 L)

A 2.85 L **B** 3.16 L **C** 0.95 L **D** 12 L

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Explain to someone at home why a nurse converts pounds to kilograms before finding a dose.